IDENTIFYING KEY INFLUENCING FACTORS FOR PREDICTING STUDENT EXAM SCORES

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Background

Student failure and dropout rates in education have adverse effects on both students and institutions

Educational outcomes are also strongly influenced by factors such as socioeconomic status, access to academic support, and the presence of learning disabilities

Understanding and predicting these factors is essential for improving student outcomes and increasing retention and success in educational institutions

Early prediction of exam scores can enable targeted interventions for at-risk students, allowing institutions to allocate resources more effectively and improve retention rates

Research Questions and Hypotheses

Hypotheses:

1. Higher levels of parental involvement and parental education will significantly contribute to improved academic performance.

2. Increased time of studying and attendance rate will be the strongest features associated with higher exam scores.

Research Questions:

1. Can machine learning models accurately predict a student's exam score based on academic, behavioral, demographic, and socioeconomic factors?

2. Which features are most influential in predicting student academic performance?

Dataset Overview

Source: Student Academic Performance Dataset (Kaggle)

Structured tabular data covering academic, behavioral, familial, and socioeconomic attributes

6,607 student records, 20 variables

Dependent variable: *Exam_Score*

Range: 55–100

Mean: 67.24

Independent Variables

Academic: *Hours_Studied, Previous_Scores, Tutoring_Sessions, Attendance, Teacher_Quality*

Behavioral: *Motivation_Level, Extracurricular_Activities, Physical_Activity, Sleep_Hours, Peer_Influence*

Demographic: *Gender, Distance_from_Home, School_Type, Learning_Disabilities, Parental_Involvement*

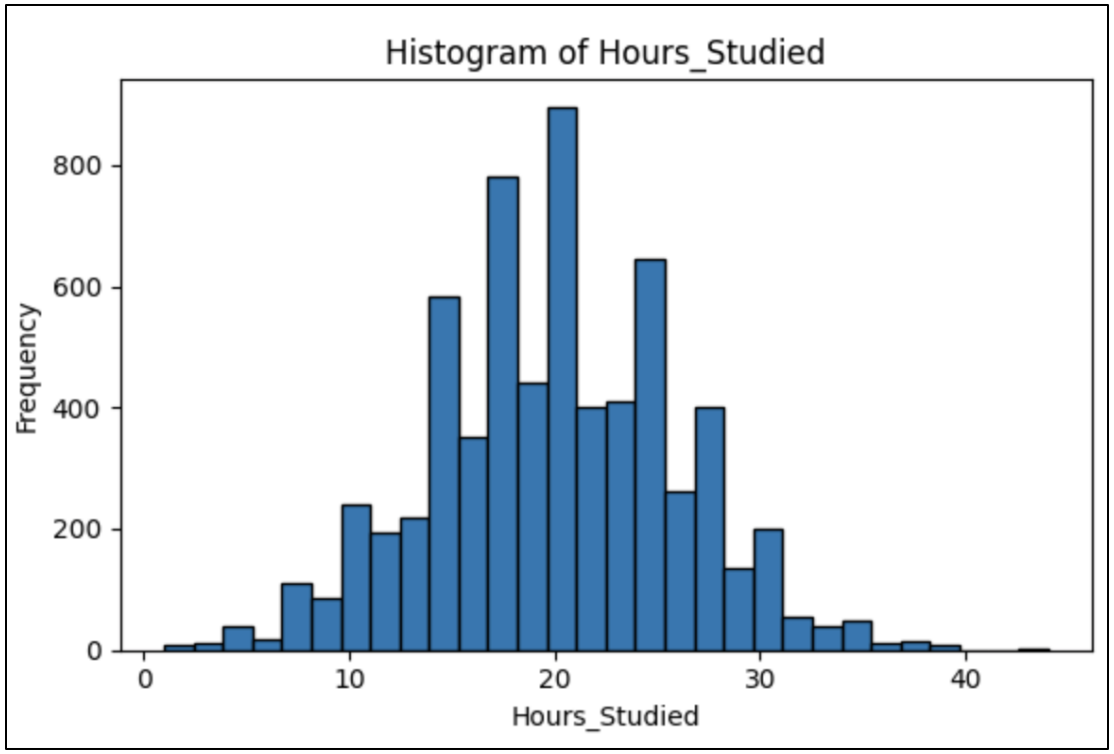
Socioeconomic: *Family_Income, Parental_Education_Level, Access_to_Resources, Internet_Access*

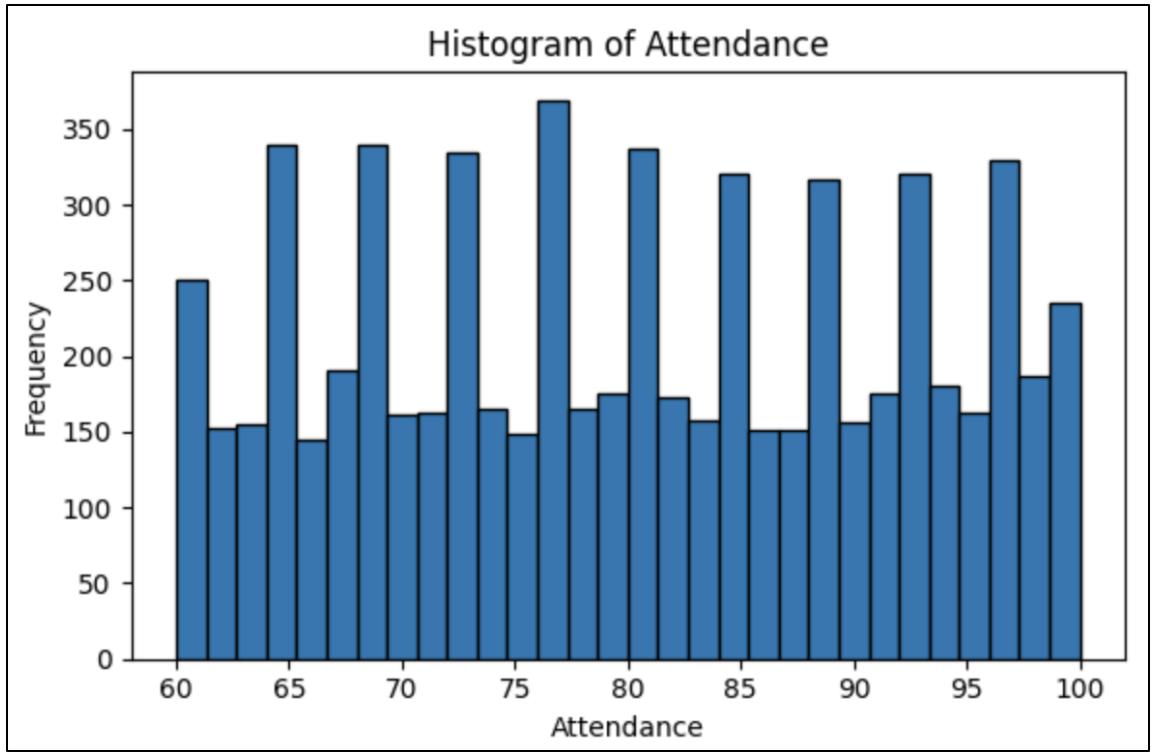
Exploratory Data Analysis (EDA)

- Began by exploring variable types, shape, missing values, and skew
- Used visualizations, summaries, and correlational analysis
- Missing values present: *Teacher_Quality* (78), *Parental_Education_Level* (90), *Distance_from_Home* (67)
- Mix of continuous (int64) and categorical (object) variables

```
RangeIndex: 6607 entries, 0 to 6606
Data columns (total 20 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Hours_Studied               6607 non-null   int64
 1   Attendance                  6607 non-null   int64
 2   Parental_Involvement        6607 non-null   object
 3   Access_to_Resources         6607 non-null   object
 4   Extracurricular_Activities  6607 non-null   object
 5   Sleep_Hours                 6607 non-null   int64
 6   Previous_Scores             6607 non-null   int64
 7   Motivation_Level            6607 non-null   object
 8   Internet_Access             6607 non-null   object
 9   Tutoring_Sessions           6607 non-null   int64
 10  Family_Income               6607 non-null   object
 11  Teacher_Quality             6529 non-null   object
 12  School_Type                 6607 non-null   object
 13  Peer_Influence              6607 non-null   object
 14  Physical_Activity           6607 non-null   int64
 15  Learning_Disabilities       6607 non-null   object
 16  Parental_Education_Level    6517 non-null   object
 17  Distance_from_Home          6540 non-null   object
 18  Gender                      6607 non-null   object
 19  Exam_Score                  6607 non-null   int64
```

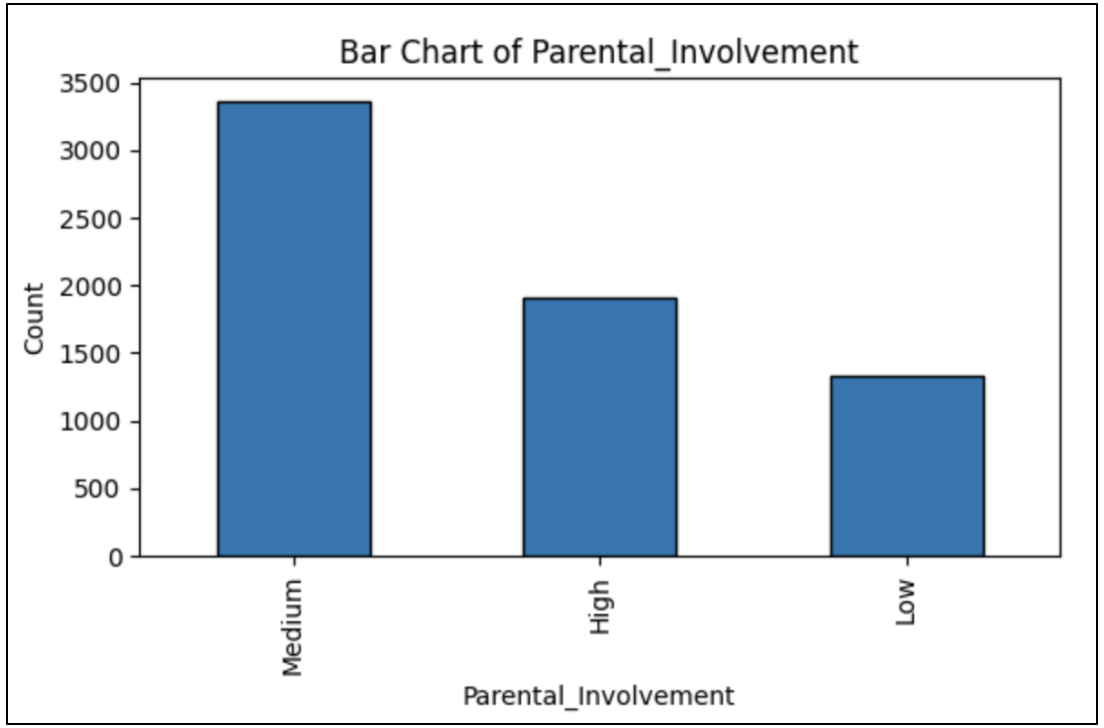
EDA Histograms

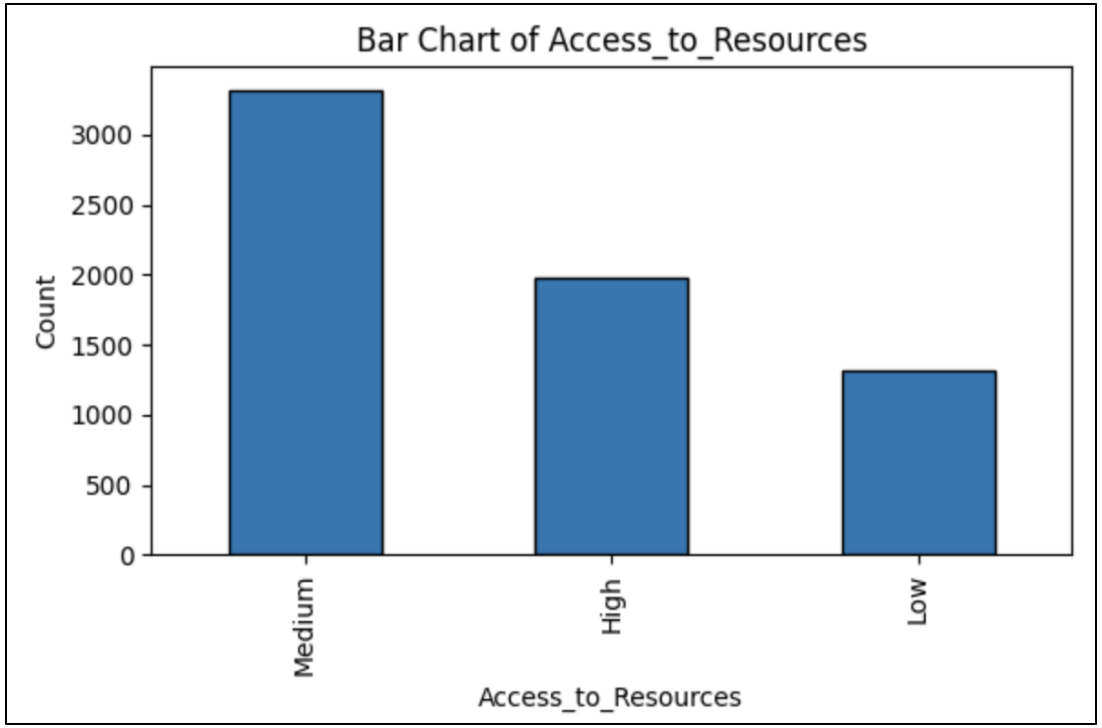




- Histograms were used to analyze the distributions of continuous variables
- Most independent variables showed relatively normal distributions, like in *Hours_Studied*
- Some were more unique like *Attendance,* with a roughly uniform distribution
- No severely skewed independent variables were found

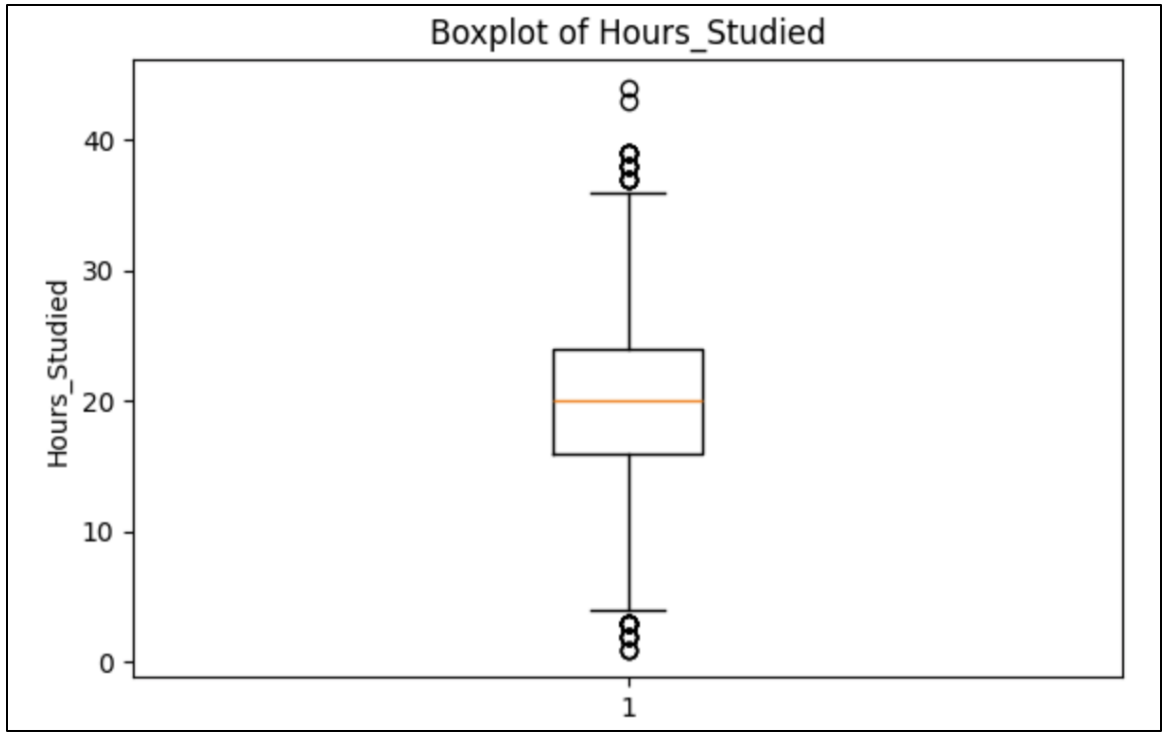
EDA Bar Charts

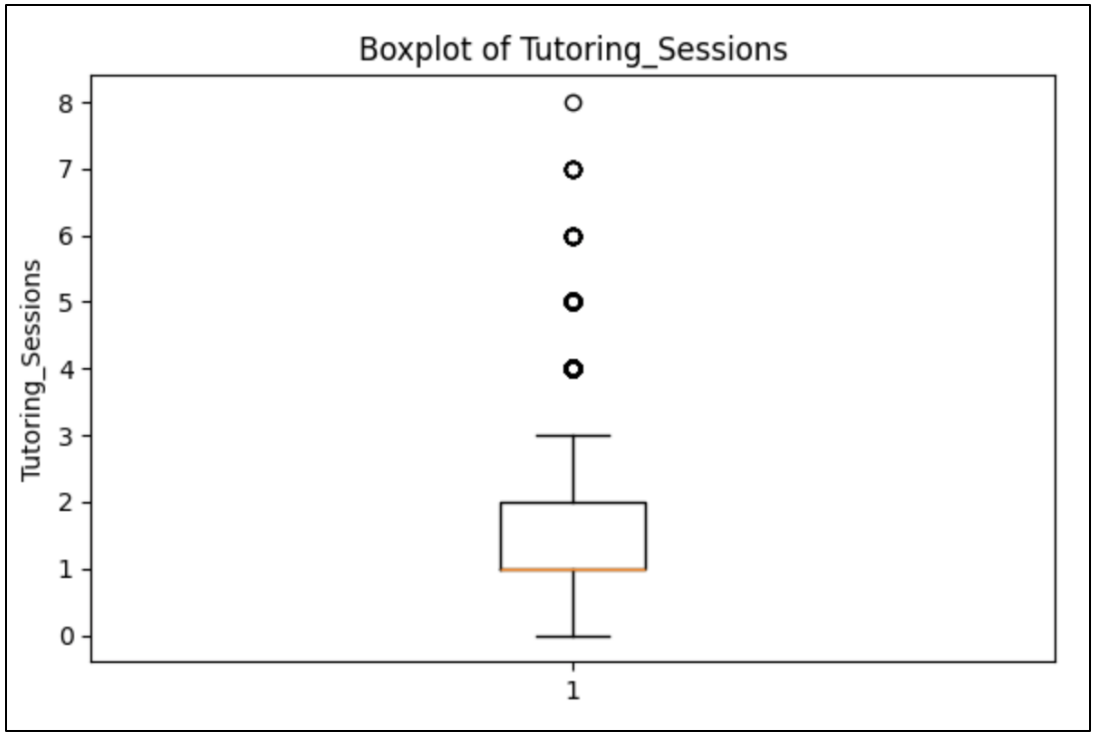




- Bar charts were used to explore the distribution of categorical variables
- Both *Parental_Involvement* and *Access_to_Resources* skewed towards medium
- This was seen universally across independent categorical variables

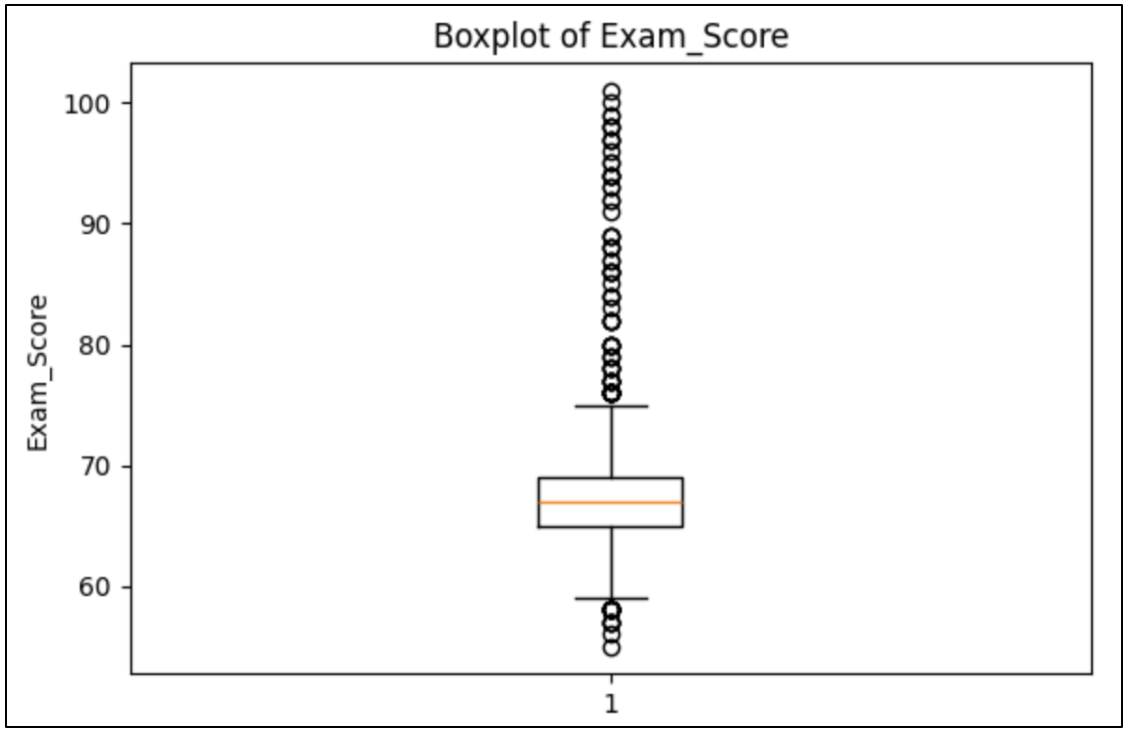
EDA Box Plots

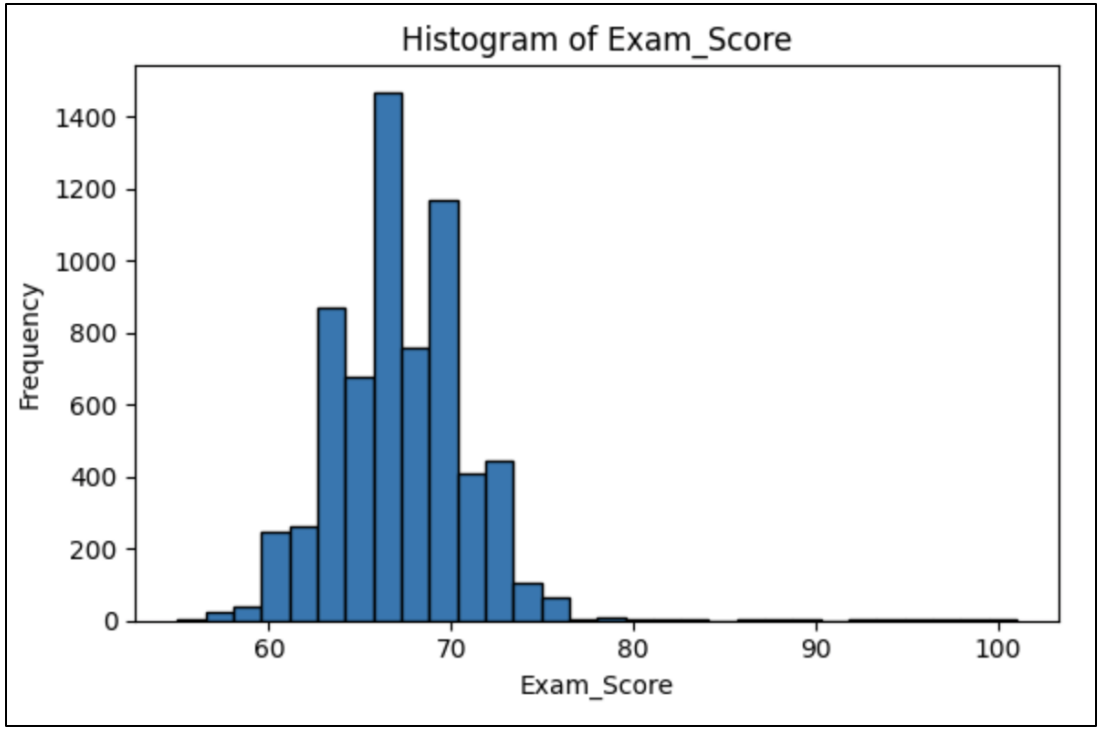




- The only notable outliers were seen in *Hours_Studied* and *Tutoring_Sessions*
- *Hours_Studied* has a median of 20 hours with a few high outliers above 35 hours per week
- *Tutoring_Sessions* is positively skewed, as most students attend 0–3 sessions, with outliers up to 8
- Outliers were retained, representing realistic student behaviors rather than data errors

EDA Dependent Variable

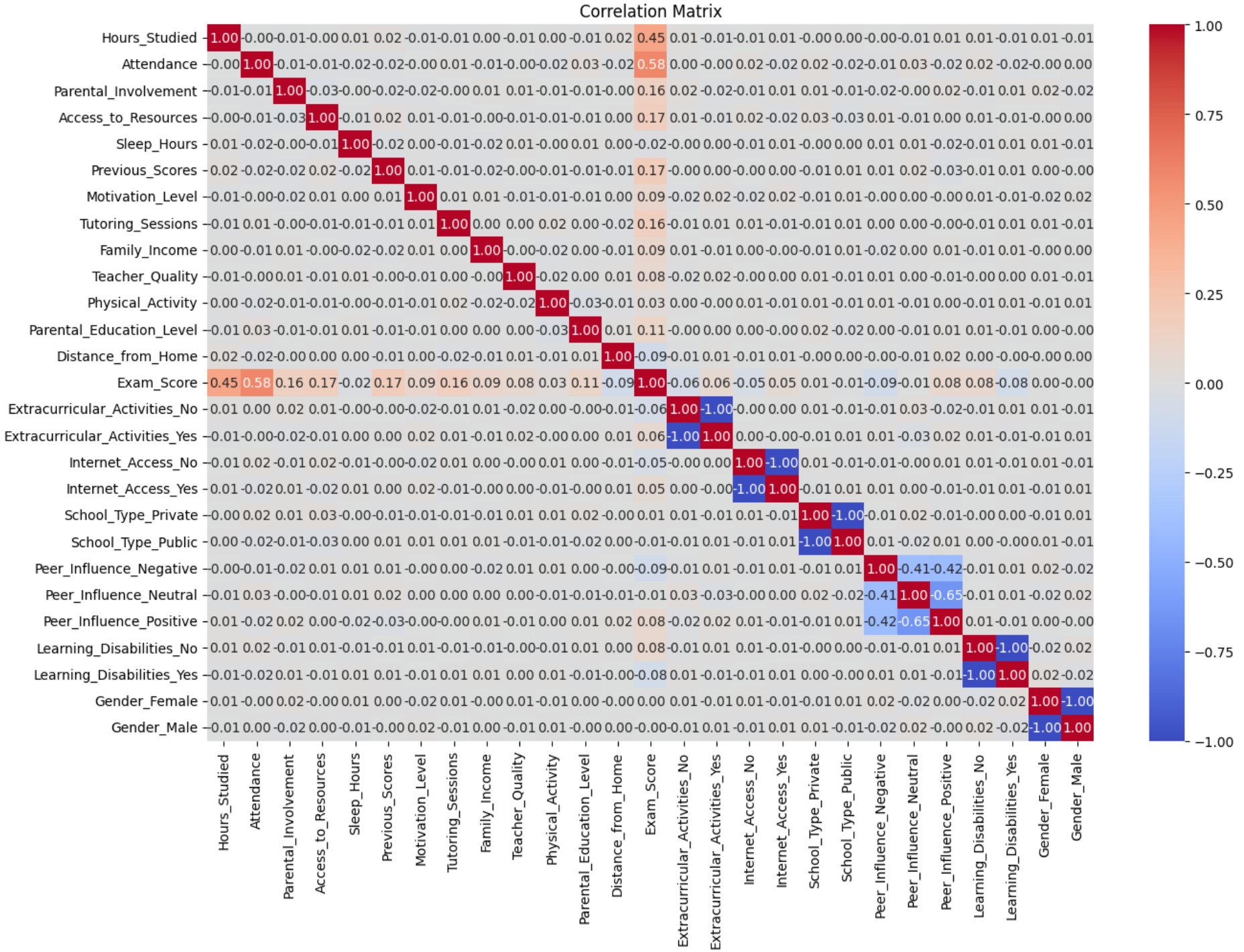




- *Exam_Score* shows a slight positive skew, concentrated between 60–75 with a median of about 67
- High outliers extend to 100, representing exceptional student performance
- Transformation was not deemed necessary

Correlational Analysis

- *Attendance*, r=0.58, and *Hours_Studied* (r=0.45) showed the strongest correlations with *Exam_Score*
- Most other features showed weak correlations with the target
- No notable multicollinearity detected among independent variables
- We decided to retain all features for our baseline model



Preprocessing

- Less than 4% of the rows contained nulls, so 229 null rows dropped
- Ordinal Variables were encoded by mapping
- Non-hierarchical categorical variables encoded with dummy variables
- New dimensions: (6378 rows x 27 columns)
- 80/20 train-test split applied with sklearn
- StandardScaler was applied to normalize features

Hours_Studied	0
Attendance	0
Parental_Involvement	0
Access_to_Resources	0
Extracurricular_Activities	0
Sleep_Hours	0
Previous_Scores	0
Motivation_Level	0
Internet_Access	0
Tutoring_Sessions	0
Family_Income	0
Teacher_Quality	78
School_Type	0
Peer_Influence	0
Physical_Activity	0
Learning_Disabilities	0
Parental_Education_Level	90
Distance_from_Home	67
Gender	0
Exam_Score	0

Baseline Linear Model

- Linear regression was selected as our baseline for simplicity, interpretability, and because *Exam_Score* is continuous
- Test R^2 was measured at 0.7324, explaining over 73% of *Exam_Score* variance
- Train and Test R^2 had minimal difference, confirming no overfitting
- RMSE was about 2 points on a 100-point scale, showing strong baseline performance

Metric	Value
Train R^2	0.7185
Mean Squared Error	4.1581
Root Mean Squared Error	2.0391
Test R^2	0.7324

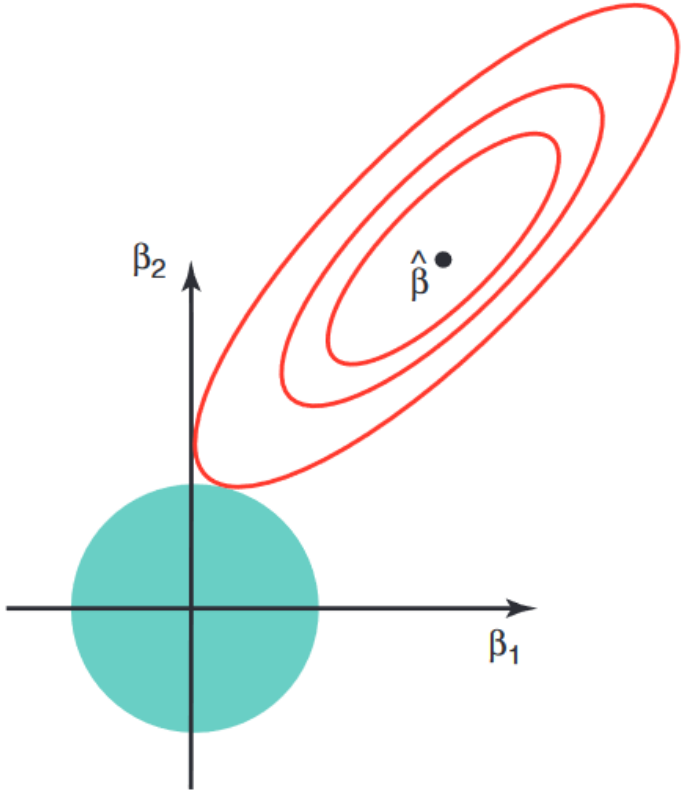
Feature Selection and Regression

- Performed stepwise feature selection with sklearn's SequentialFeatureSelector, which iteratively adds features that most improve the 5-fold CV score
- 13 features selected: *'Hours_Studied', 'Attendance', 'Parental_Involvement', 'Access_to_Resources', 'Previous_Scores', 'Motivation_Level', 'Tutoring_Sessions', 'Family_Income', 'Teacher_Quality', 'Parental_Education_Level', 'Distance_from_Home', 'Extracurricular_Activities_No', 'Peer_Influence_Positive'*
- R^2 decreased and RMSE increased, implying the model's performance got worse
- Dropped features were individually weak but collectively added meaning

Metric	Value
Train R^2	0.7049
Mean Squared Error	4.3089
Root Mean Squared Error	2.0758
Test R^2	0.7227

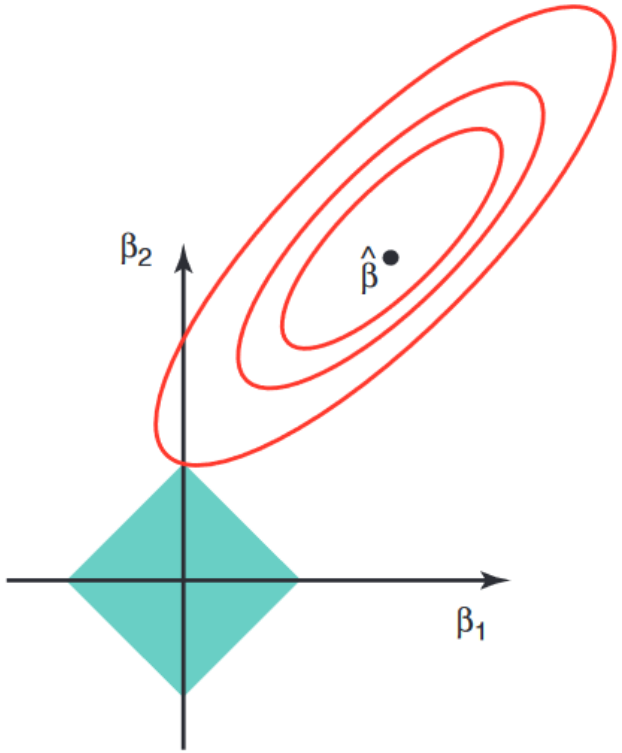
Ridge Regularized Regression

- Ridge was used to test whether regularization could improve on the baseline by reducing potential overfitting
- Adds an L2 penalty based on the square of coefficients, shrinking coefficients toward 0, without eliminating them
- Used RidgeCV to find the optimal alpha with 5-fold cross-validation
  - [0.01, 0.1, 1, 5 10, 15, 20, 25, 50, 100] -> 20
- Model results were nearly identical to baseline, confirming no overfitting present



Metric	Value
Train R^2	0.7185
Mean Squared Error	4.1577
Root Mean Squared Error	2.0391
Test R^2	0.7324

Lasso Regularized Regression

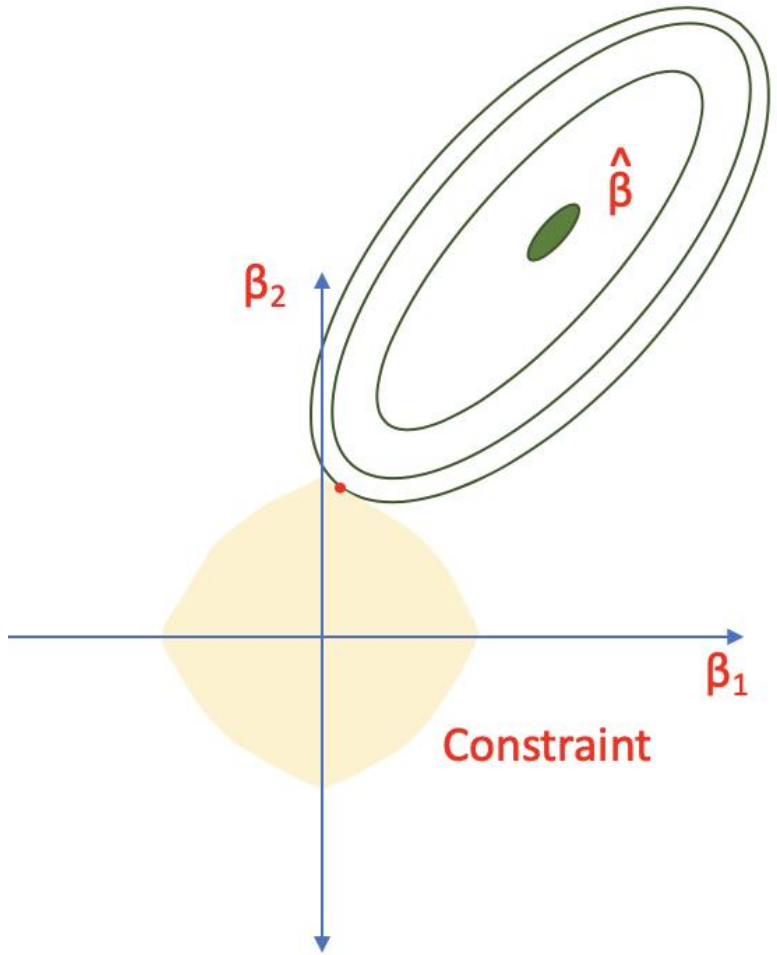


- Lasso adds an L1 penalty based on the absolute value of coefficients, shrinking coefficients to exactly 0
- Used LassoCV to find the optimal alpha with 5-fold cross-validation
  - [0.0001, 0.0001, 0.001, 0.005, 0.01, 0.1, 1] -> 0.005
  - The small alpha indicates minimal regularization is necessary, consistent with other results
- *Peer_Influence_Neutral* was the only variable dropped, showing all other features contributed to the model
- This was our best model created, with a slightly higher Test R^2 and slightly lower RMSE

Metric	Value
Train R^2	0.7184
Mean Squared Error	4.1567
Root Mean Squared Error	2.0388
Test R^2	0.7325

Elastic Net Regression

- Elastic net is a combination of Lasso(L1) and Ridge(L2) penalties, controlled by the L1 ratio
- Optimal hyperparameters were again selected with 5-fold cross validation:
  - Alpha (strength of penalties) - [0.0001, 0.0001, 0.001,
    0.005, 0.01, 0.1, 1, 5, 10, 15, 20, 25, 50, 100] -> 0.005
  - L1 Ratio – The mix between Lasso and ridge - [0.1, 0.3,
    0.5, 0.7, 0.9, 0.95, 0.99, 1] -> 1
- This model produced the exact same results as Lasso, supported by the L1 ratio of 1.0
  - Test R^2 = 0.7325, RMSE = 2.0388
  - *Peer_Influence_Neutral* dropped
  - Ridge penalty added no value



Metric	Value
Train R^2	0.7184
Mean Squared Error	4.1567
Root Mean Squared Error	2.0388
Test R^2	0.7325

Random Forest

- Random Forest is an ensemble method that builds many randomized decision trees, reducing variance through averaging
- We wanted to see if a nonlinear model could improve on the linear baseline
- GridSearchCV with 5-fold CV tested 648 parameter combinations across n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features, and bootstrap
- The best parameters found were max_depth=10, max_features=0.3, n_estimators=500, min_samples_leaf=12, min_samples_split=20, bootstrap=False
- As seen with the notable R^2 gap, there was overfitting with this model

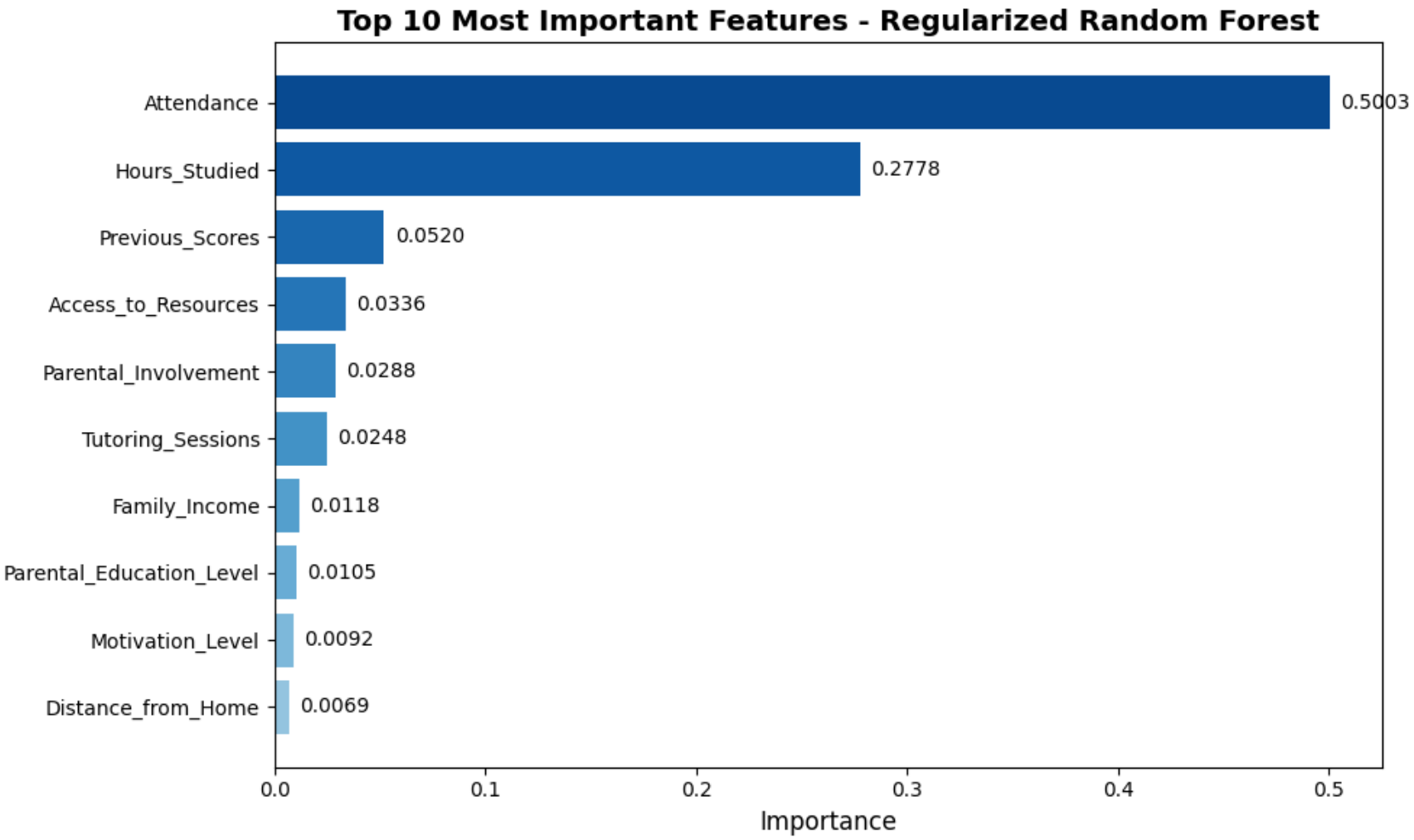
```
param_grid_2 = {
    'n_estimators': [300, 400, 500],
    'max_depth': [6, 8, 10],
    'min_samples_split': [20, 25, 30, 35],
    'min_samples_leaf': [12, 15, 20],
    'max_features': ['sqrt', 0.2, 0.3],
    'bootstrap': [True, False]
}

grid_search_2 = GridSearchCV(
    estimator=rf_2,
    param_grid=param_grid_2,
    cv=5,
    scoring='r2',
    verbose=1,
    n_jobs=-1
)
```

Metric	Value
Train R^2	0.7235
Mean Squared Error	5.5687
Root Mean Squared Error	2.3598
Test R^2	0.6416
R^2 Gap	0.0819

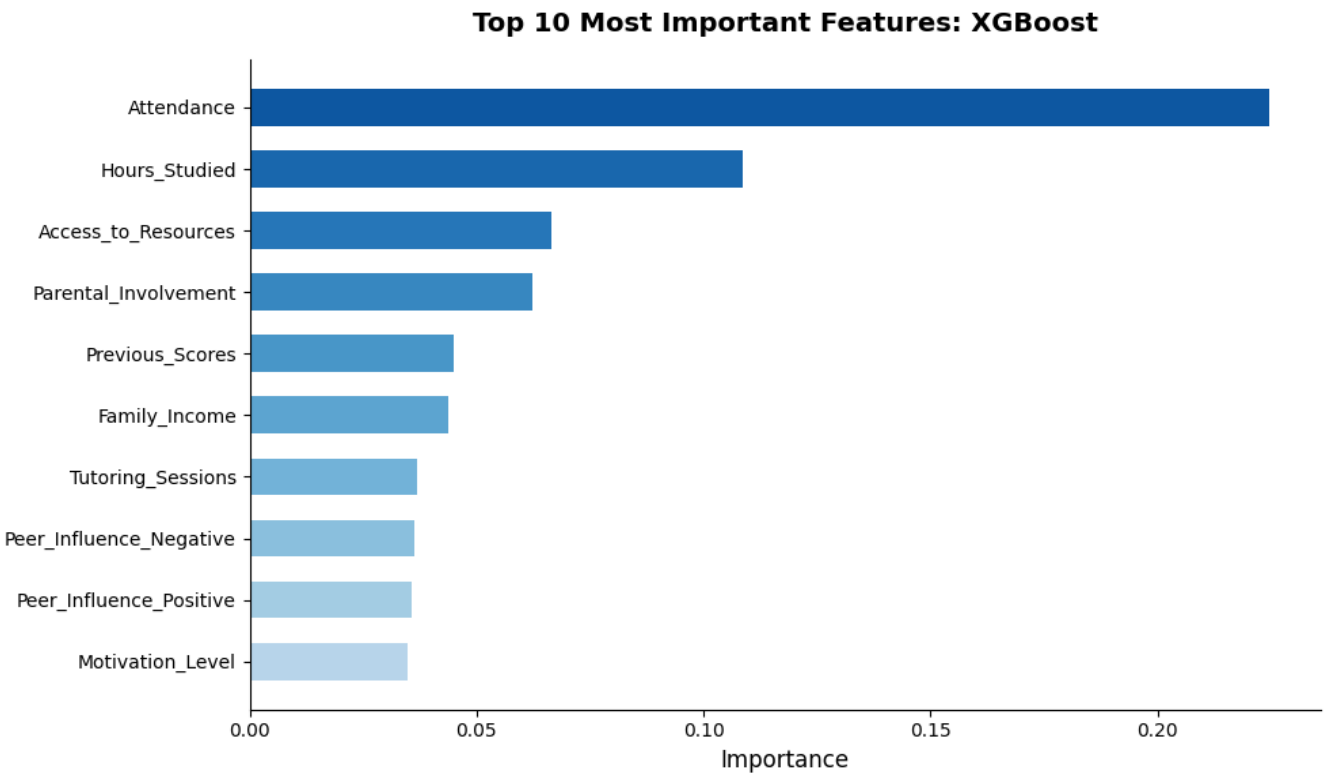
Random Forest Feature Importance

- *Attendance* and *Hours_Studied* account for nearly 78% of total feature importance
- This supports our hypothesis that study time and attendance are the strongest predictors
- Socioeconomic factors appear but with much lower importance



XGBoost

- Gradient boosting builds strong predictive model by consolidating weaker trees sequentially, correcting previous errors
- Used MSE as loss function; penalizes large prediction errors during training more heavily
- Tuned across 3 runs to reduce overfitting and improve generalization
- Top predictors: *Attendance* (0.22), *Hours_Studied* (0.11), *Access_To_Resources* (0.07), *Parental_Involvement* (0.06)



Run	Change	R^2
1	max_depth=6	0.6828
2	max_depth=4	0.7093
3	min_child_weight=3	0.7108

XGBoost Results and Interpretation

- XGBoost explains 71% of variance in exam scores; predictions within less than 1 exam point on average (MAE = 0.69)
- Outperforms RF, due to boosting's sequential correction
- Linear models remain top performers, suggesting linear relationships
- Tree-based models could also encounter more noise comparatively, affecting performance overall

Model	rmse	R^2
Linear/ridge/lasso/elastic net	2.04	0.732
XGBoost (tuned)	2.12	0.711
Random Forest	2.30	0.659

Conclusion

The Lasso regularized regression model was the best performing model with a Test $R^2 = 0.7325$ and RMSE = 2.04

This was a marginal improvement on our baseline linear regression model

Other regularization did not improve the baseline, showing no meaningful overfitting was present

Random Forest and XGBoost performed worse than the linear models, showing the relationships between the independent variables and *Exam_Score* is linear

Attendance and *Hours_Studied* are strongest predictors of performance

Revisiting Research Questions and Hypotheses

1

Research Question 1: Our machine learning model was able to predict a student's exam score within about 2 points

2

Research Question 2: *Attendance* and *Study_Hours* were the most influential features in predicting student academic performance

3

Hypothesis 1: Partially supported – Although parental guidance was shown to be relevant, it was not as important as other predictors

4

Hypothesis 2: Confirmed – *Attendance* and *Study_Hours* were consistently the top predictors across all models

Discussion

Our model was able to explain about 73% of the variance in Exam_Score

The unexplained 27% could be attributed to factors either difficult to quantify or not included in the dataset (Motivation, anxiety, etc.)

Based on the Random Forest and XGBoost results, it is fair to assume that the linearity of the model limits tree-based model success

Feature selected tended to hurt performance, suggesting that weaker features cumulatively had a significant impact

Limitations

Dataset may not reflect the diversity found in real-world student populations

The dataset was not completely clean, some values had to be dropped

Some features were more qualitative and subjective

Future Work

- Collect more records or students
- Add more features, such as class size, mental health, teacher experience
- Minimize subjective features to improve consistency homogeneity in reporting practices
- Translate exam scores into broader categories like graduation rate or final GPA

Citations

- 1. Berka, P., & Marek, L. (2021). Bachelor's degree student dropouts: Who tend to stay and who tend to leave? *Studies in Educational Evaluation, 70*, 100999. https://doi.org/10.1016/j.stueduc.2021.100999
- 2. Quincho Apumayta, R., Carrillo Cayllahua, J., Ccencho Pari, A., Inga Choque, V., Cárdenas Valverde, J. C., & Huamán Ataypoma, D. (2024). University Dropout: A Systematic Review of the Main Determinant Factors (2020-2024). *F1000Research*, *13*, 942. https://doi.org/10.12688/f1000research.154263.2